

Pratyush Sharma

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CORPORATE EXPERIENCE

- ◆ **Quantitative Research Intern** | *WorldQuant LLC* (May 2026 – July 2026)
 - Used in-house risk models and market-theme data for market-responsive clustering of stocks.
 - Encoded varied data types into a unified affinity matrix, enabling data type agnostic clustering.
 - Improved the intra-cluster correlation of future returns compared to standard static clustering.

RESEARCH INTERNSHIPS

- ◆ **Fine Grained Complexity** | *Prof. Adam Polak* | *Bocconi, Italy* (May 2024 – July 2024)
 - Improved on state of the art algorithm for the 3SUM problem in the pre-processed setting from $\tilde{O}(N^{11/6})$ to $O(N^{1.5} \log N)$ using Fast Fourier Transform and Hashing techniques.
 - Presently working on another research that improves time complexity of **Dynamic Range Mode**.
- ◆ **Data Compression** | *Prof. Kaushik Sinha* | *Wichita State, USA* (May 2025 – July 2025)
 - Surveyed the current state of the art algorithm for *Condensed Nearest Neighbours*.
 - Reviewed selective sampling in NN models, and the bounds of $O(\log N)$ asymptotic memory storage.

PUBLICATIONS

3SUM in Preprocessed Universes: Faster and Simpler

Pratyush Sharma, Adam Polak, Shashwat Kasliwal. Accepted at the **TheoretiCS** journal [Journal] and at the **Symposium on Simplicity in Algorithms (SOSA)'25**, New Orleans, USA [arXiv].

SCHOLASTIC ACHIEVEMENTS

- Secured 1st position in the **Spring India Citadel Terminal 2024**, a **pan India** algorithmic tower strategy game hosted by *Correlation One*, winning the Grand Prize of \$6500. (Sponsored by Citadel)
- Achieved **AIR 399 (Gen)** in **JEE Advanced 2022** and **AIR 335 (Gen)** in **JEE Mains 2022**.
- Secured 1st position in **IIT Delhi** in the first and second prelims and represented the institute at the **Integral Cup** finals, competing against other IIT's in a *MIT style* Integration Competition.
- Secured **All India Rank 10** amongst 1.5 million students and received the **KVPY – SA** scholarship.
- Secured **All India Rank 1** in **LIMIT Exam** amongst all Undergraduate students, hosted by *ISI*.
- Achieved **Oxbridge STEM Scholarship** based on past credentials in STEM related academics.
- Secured **AIR 2** in **Mathemania'25** and **AIR 9** in **Mathemania'24** hosted by *IIT Kanpur*.
- Achieved **All India Rank 1** in **ResoStart** and awarded a *sponsored 9 days* trip to **NASA**.
- Secured **All India Rank 1** in Aakash National Talent Hunt Examination (**ANTHE**).

OLYMPIAD ACHIEVEMENTS

- Secured the 3rd position amongst all the teams in the East Division & (1st in *IIT Delhi*) of the Simon Marais Maths Competition **SMMC**, received the **Best Team Prize**. (Sponsored by Jane Street/Optiver)
- Secured 2nd position in 14th **Inter IIT Quant Competition** hosted by *Ebullient Securities*.
- Secured 6th position in 13th **Inter IIT Maths Competition** hosted by *Chennai Maths Institute*.
- Secured **International Rank 4** in **Gamma Maths Olympiad (IGMO)**. (Sponsored by Jane Street)
- Selected to represent and lead the **IIT Delhi** contingent at the **Al-Khorezmi IMO (Uzbekistan)**
- Qualified and Attended **IJSO Junior Science National Camp** (Top 35 in India) in 9th std.
- Qualified for **IOAA Astronomy National Camp** (Top 35 in India) in 10th, 11th & 12th std.
- Qualified for **IPHO Physics National Camp** (Top 35 in India) in 12th std.
- Secured **AIR 7** in **Srinivasa Ramanujan Mathematical Competition (SRMC)** hosted by *IIT Bombay*.

PROJECTS

- **Inter IIT Quant** | *Ebullient Securities* | *Silver Medal* (December 2025)
 - Engineered alpha signals via crossovers of **statistical indicators** for systematic strategies.
 - Modeled time-series price data as sequential context for **sequence-tagging** classification.
 - Built orthogonal alphas from **SHAP-selected features**, integrated with portfolio optimization.
 - Segmented market regimes and volatility via unsupervised **HMM/k-means** clustering.
- **Actor Success Prediction:** | *Prof. Niladri Chatterjee* | *ML* (October 2024)
 - Self implemented ML models (*KNN, RF, LR*) with robust preprocessing and tuning to classify data.
 - Boosted accuracy using **NLP (TF-IDF, NER)** and feature engineering on the dataset.
- **AQI Prediction:** | *Prof. Niladri Chatterjee* | *ML* (December 2024)
 - Developed a **distributed learning** framework supporting major models for distributed edge sensors.
 - Enabled local model training with data privacy, improving prediction accuracy across networks.
- **Automated Resume Analyzer:** | *Eightfold AI* | *Hackathon Winner* (March 2024)
 - Used Hugging Face's **BERT** for word embeddings, and *cosine similarity* for relevance to query.
 - Utilized **PyMuPDF** for OCR text extraction from CV and highlighting the points from the PDF.
- **Search Engine with Query Processor:** | *Prof. Rahul Garg* | *DSA* (October 2023)
 - Implemented a **query processor** inputting a corpus and query to give the most relevant paragraphs.
 - Linked the query processor to ChatGPT using the API tokens to utilise the API keys in C++.
 - Used **spaCy** and **TF-IDF** to increase the weight of chosen words and filter out stop words.
- **Expression Evaluator and Compiler:** | *Prof. Subodh Sharma* | *DSA* (November 2023)
 - Developed a **language compiler (E++)** for equation solving using *self-implemented AVL and BST*.
 - Generation of code for parse trees in **Targ** (target machine language) to evaluate expressions.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, MATLAB, Mathematica, Brainf**k.
- **Software & Development:** L^AT_EX, Git, Numpy, SciKit, Pandas, Matplotlib, PyTorch, Manim.

EXTRACURRICULARS

- Featured on **Jane Street** Puzzles website for solving the puzzle of the month *multiple* times.
- **Coordinator MathSoc (IITD):** Led a 10-student team for placement preparation (*May '23 – May '25*)
- Previously the **Captain** of the **IIT Hostel Chess** team. Secured 1st in General Championship 2026.
- **TRYST, IIT Delhi 2023:**
 - 1st in **Codehunt**, a Gamified Competitive Programming contest conducted by *ACES ACM*.
 - 2nd in **GenoQuest**, a competition of synthetic and computational biology conducted by *iGEM*.
 - 3rd in **Integration Bee**, a competition of college integration questions conducted by *MathSoc*.
 - 5th in **Formula Brawl** a competition of physics formulas quiz conducted by *Astronomy Club*.
- **TRYST, IIT Delhi 2024:**
 - 2nd in **GenoQuest**, a competition of synthetic and computational biology conducted by *iGEM*.
 - 3rd in **Integration Bee**, a competition of college integration questions conducted by *MathSoc*.
- **TRYST, IIT Delhi 2025:**
 - 1st in **Number Games**, a competition of calculation and mental maths conducted by *MathSoc*.
 - 1st in **Integration Bee**, a competition of college integration questions conducted by *MathSoc*.
 - 3rd in **CodeDecode**, a new twist on CP by reverse engineering code conducted by *ANCC*.
- **TRYST, IIT Delhi 2026:**
 - 2nd in **Stratezenith**, a competition of multiple game theory rounds conducted by *IGTS*.
 - 1st in **Clash Royale** competition hosted by *Sporttech* among 128+ other participants.